

From Behavior Cloning to Offline Reinforcement Learning

Learning Better Decisions from Logged Data

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From Prediction to Decision

Prediction	Decision
What will happen?	What should we do?
Truck arrival time	Which truck to unload next
Grain quality	Where to route the grain
Train delay	Which loading schedule to use

Machine learning describes the operation. Decision-making changes it.

Reinforcement learning is a framework for learning such decisions.

Why Decisions Are Different



- ▶ Supervised learning learns from a fixed dataset.
- ▶ Decision-making selects actions rather than making predictions.
- ▶ Those actions influence future states and therefore the data seen later.

**Prediction assumes a mostly stationary data source.
Decisions change the state distribution the model will face next.**

How Have We Solved Decision Problems?

**Rules and
expert systems**

Encode operational
knowledge directly.

**Optimization
and control**

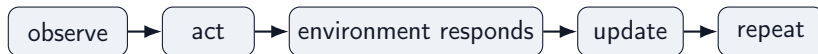
Model the system
and compute
good actions.

**Reinforcement
learning**

Improve decisions
using experience
and feedback.

These are not mutually exclusive: practical systems often combine expert knowledge, control, optimization, and learned policies.

The Traditional Reinforcement Learning Paradigm



Classical reinforcement learning:

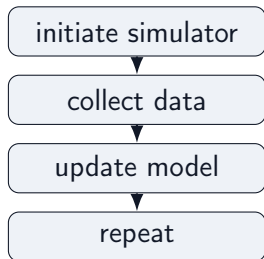
- ▶ learns from consequences rather than labeled targets,
- ▶ balances exploration and exploitation,
- ▶ optimizes long-term reward.

For much of its history, reinforcement learning was framed as learning through continual interaction.

In practice, this usually means learning in a simulator before deployment.

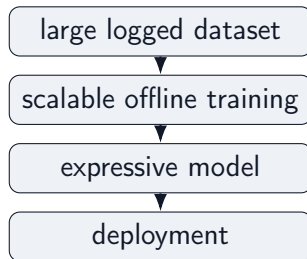
A Data-Centric View of Reinforcement Learning

Interaction-centric RL



Improve through repeated interaction.

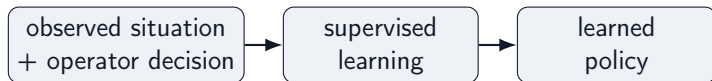
Data-centric ML



Transformers and foundation-model-style scaling benefit from large static datasets.

What can we learn from the interaction data we already have?

Start Simple: Behavior Cloning



- ▶ Observe what an operator or existing controller did.
- ▶ Train a model to predict the same action in similar situations.
- ▶ Deploy the trained model as a learned “policy”.

Behavior cloning is fully offline and requires no simulator or reward model.

Can we do more than learn based on operator data? Can we prefer decisions that led to better long-term outcomes?

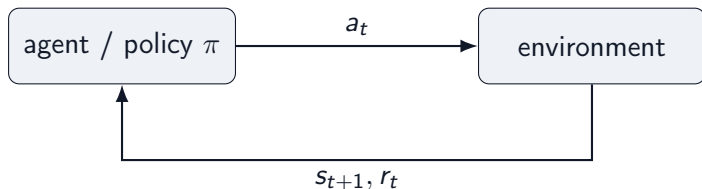
To answer that, we first need a language for decision problems: states, actions, and policies.

States, Actions, and Policies

Object	Grain-terminal example
state s_t	current queue, silo inventory, grain quality, train progress
action a_t	unload truck, choose pit, route conveyor, assign silo, fill car
policy $\pi(a \mid s)$	rule or model that chooses actions from states
next state s_{t+1}	the facility after the action has changed it

A decision problem starts with states, actions, and a policy that maps situations to decisions.

Agent-Environment Loop



- ▶ **Agent:** chooses the next action from the current state.
- ▶ **Environment:** the facility or simulation that returns the next state and reward.
- ▶ **Key contrast:** supervised learning predicts from fixed data; RL actions create the next data.

Imitation Learning: Turn Decisions Into Supervised Learning

First idea: collect observed behavior and train a policy to predict what the operator would do.

Historical demonstrations come from a **behavior policy** π_b , e.g. human terminal operators:

$$\mathcal{D} = \{(s_t, a_t)\}, \quad a_t \sim \pi_b(a \mid s_t)$$

Train π_θ as a supervised action predictor:

Continuous actions

steering angle, conveyor speed

Discrete actions

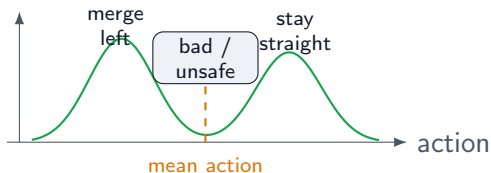
route choice, silo assignment

$$\min_{\theta} \mathbb{E}_{\mathcal{D}} [\|a - \pi_{\theta}(s)\|^2]$$

$$\min_{\theta} -\mathbb{E}_{\mathcal{D}} [\log \pi_{\theta}(a \mid s)]$$

Attractive because it is fully offline, simple to train, and does not require defining a reward.

Pitfall 1: Averaging Demonstrations Can Fail

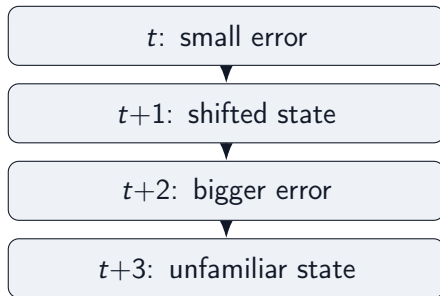


Consider imitation data for self-driving cars:

- ▶ in similar observed states, driver 1 stays in lane
- ▶ driver 2 changes lanes
- ▶ squared-error regression learns the conditional mean action

Problem: the mean action can lie between valid modes: a poor command that no expert demonstrated.

Pitfall 2: Errors Accumulate Over Time



A policy trained only on expert states may not know how to recover from states caused by its own mistakes.

- → Distribution shift: deployment samples from the learned policy's state distribution, not the expert's.

Problem: one-step prediction errors can look small while long rollouts become unstable.

Pitfall 3: Imitation Ignores Outcomes

Behavior cloning asks only:

“What action was taken in this situation?”

It does not ask:

“Which action led to the best long-term outcome?”

Every logged decision becomes a training example. The quality of the outcome does not change how strongly it is learned.

Problem: behavior cloning reproduces past decisions, but it cannot prefer decisions with better long-term consequences.

Rewards Measure the Consequences of Decisions

A reward turns delayed product outcomes into feedback for decisions:

$$r_t = r(s_t, a_t, s_{t+1})$$

For grain loading, rewards can combine:

- ▶ train loading progress and throughput
- ▶ contract quality satisfaction
- ▶ penalties for rehandling, congestion, and delay

Rewards tell the learner what outcomes matter, not just what actions were taken.

Long-Term Consequences

A good decision is one that leads to *high future reward*, not just a high immediate reward.

$$\text{Future Reward} = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$$

where $0 < \gamma < 1$ discounts delayed rewards.

A value function predicts this quantity before making a decision:

$$V(s) \approx \text{future reward, starting from state } s$$

$$Q(s, a) \approx \text{future reward, starting from state } s \text{ and taking action } a$$

Main takeaway

V and Q quantify how good it is to be in a state and to take an action. They are central objects in RL.

Learning the Value Function

Learning the Q -function is an important step in nearly all of RL.

To keep things simple, Q satisfies a recursive relation:

long-term value = expected current reward + discounted future value.

$$Q(s_t, a_t) \approx r_t + \gamma V(s_{t+1})$$

This recursion can be solved with exact or approximate dynamic programming.

We will skip those details and use the equation only as the basic intuition behind value learning.

Three Influential Ideas in Offline Reinforcement Learning

Complementary approaches to improve behavior cloning:

1. **Reweight the logged data**

Advantage-Weighted Regression (AWR): copy successful decisions more strongly.

2. **Learn conservative value estimates**

Conservative Q-Learning (CQL): be pessimistic where evidence is weak.

3. **Estimate values only from supported actions**

Implicit Q-Learning (IQL): avoid maximizing over unsupported actions altogether.

Three Common RL Families

Family	Basic idea
Value-based RL	Learn $Q(s, a)$, act greedily
Policy gradient	Directly improve $\pi(a s)$
Actor-critic	Learn both policy and value

- ▶ Q-learning is conceptually simple, but is most natural when actions are discrete.
- ▶ Policy gradients apply more generally, but require expensive rollouts.
- ▶ Actor-critic methods combine both ideas: they learn a policy and a value estimate, and work with discrete or continuous actions.

Generic Actor-Critic Offline RL

Most modern actor-critic Offline RL methods separate two steps:

estimate Q, V



improve policy

The three ideas intervene at different points:

- ▶ **AWR** mainly changes the improve-policy step.
- ▶ **CQL** and **IQL** focus primarily on value estimation.

Policy updates need values; methods differ in how those values are made reliable.

Problem: Not All Demonstrations Are Equally Useful

Behavior cloning treats every logged decision equally.

But logged data often contains a mixture:

- ▶ strong operator decisions,
- ▶ reasonable routine decisions,
- ▶ choices that led to delay, rehandling, or poor downstream outcomes.

Question

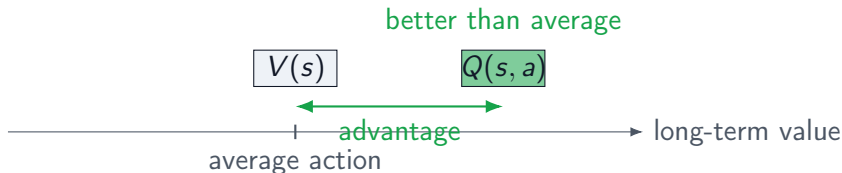
Can we learn more from the better demonstrations?

Core Idea: Advantage

Compare the logged action to what is typical in the same state:

$$A(s, a) = Q(s, a) - V(s).$$

- ▶ $Q(s, a)$ measures how good it is to take action a in state s .
- ▶ $V(s)$ measures how good the state is on average.
- ▶ Their difference says how much better this action is than expected.



This difference is called the **advantage**.

Advantage-Weighted Regression (AWR)

Behavior cloning solves:

$$\min_{\theta} \sum_i \ell(\pi_{\theta}(s_i), a_i)$$

AWR keeps the same supervised-learning loss, but weights each example:

$$\min_{\theta} \sum_i \exp(\beta A(s_i, a_i)) \ell(\pi_{\theta}(s_i), a_i) .$$

Key point

The only change is the weighting. Everything else remains supervised learning.

Next question: Where do reliable estimates of Q and V come from?

Problem: Offline Values Can Be Overconfident

To compute useful weights, we need value estimates.

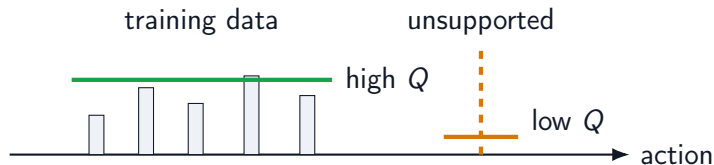
But value estimation is harder offline:

- ▶ online learning can test actions and correct bad guesses,
- ▶ offline learning cannot collect new feedback,
- ▶ unsupported actions may receive overly optimistic predicted values.

Question

How can we learn values without trusting predictions where the data is weak?

Core Idea: Conservative Value Learning



$$\mathcal{L}_{\text{CQL}} = \underbrace{\mathcal{L}_{\text{Bellman}}}_{\text{fit observed transitions}} + \alpha \left(\underbrace{\mathbb{E}_{a \sim \mu} Q(s, a)}_{\text{push down candidate actions}} - \underbrace{\mathbb{E}_{(s, a) \sim \mathcal{D}} Q(s, a)}_{\text{keep logged actions high}} \right).$$

Main takeaway

CQL changes only the value-learning objective: fit the data, but penalize high Q-values for unsupported actions.

Another Design Philosophy: Avoid the Max

Conservative value learning is one way to obtain reliable offline values.

Another family asks whether we can avoid unsupported maximization in the first place.

Many value-based methods rely on searching over actions:

$$\max_a Q(s, a)$$

Offline, this search can expose value errors in weakly supported regions.

Alternative question

Can we estimate values without maximizing over unsupported actions at all?

A Different Idea: Never Evaluate Unseen Actions

CQL keeps the maximization, but makes unsupported actions pessimistic.

Can we avoid the maximization entirely?

Standard Q-learning uses the target:

$$Q(s, a) \leftarrow r + \gamma \max_{a'} Q(s', a').$$

The problem is that the **maximization** evaluates actions that never appeared in the dataset, creating extrapolation errors.

Main takeaway

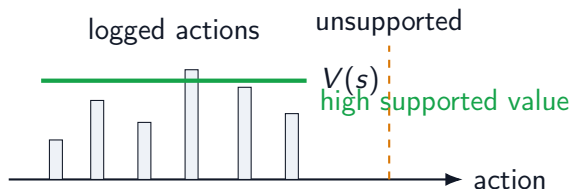
IQL's key idea: never evaluate actions that are not in the dataset.

Learn the Best Supported Value

Instead of computing $\max_a Q(s, a)$, IQL learns a state value:

$$V(s) \approx \text{upper expectile of } Q(s, a) \text{ for logged actions } a.$$

- ▶ mean: average action
- ▶ maximum: possibly unsupported action
- ▶ upper expectile: high but supported action



Expectile regression is the tool used to estimate this high supported value.

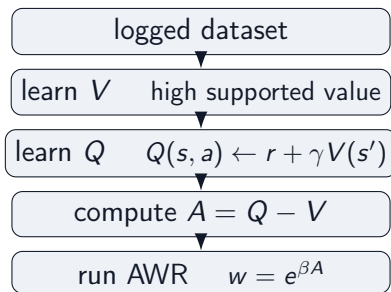
Then learn Q with:

$$Q(s, a) \leftarrow r + \gamma V(s').$$

Notice: $\max_a Q(s, a)$ has disappeared.

Policy Extraction: Reuse AWR

Once $V(s)$ and $Q(s, a)$ have been learned, the policy update is familiar.



Main takeaway

IQ-L changes value estimation, not policy optimization: $V \rightarrow Q \rightarrow A \rightarrow \pi$

Complementary Issue: Several Good Actions



The first three ideas focused on value estimation.

IDQL addresses a different issue: representing policies when several actions are equally good.

Diffusion Policies Represent Richer Action Distributions



A diffusion policy can sample several plausible actions for the same state.

IDQL combines IQL-style value learning with diffusion-based policy extraction.

Summary: What Changes Across Methods?

Method	Main contribution	Role
AWR	Advantage-weighted policy improvement	Policy update
CQL	Conservative value estimation	Value learning
IQL	In-sample value estimation	Value learning

Main message

Modern Offline RL largely consists of estimating reliable values from logged data, then improving the policy using those values.

Different methods primarily differ in how reliable values are obtained, not in the overall objective.

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